Assignment 5 - Command Line File Management in Linux

The objective of this lab is to gain hands-on experience with command-line file management in Linux. Through this lab, you will learn essential commands for creating, modifying, and navigating files and directories using the command line.

Access a Linux terminal for the following exercises.

Questions for this assignment

1. Please provide a practical explanation of Basic File and Directory Operations.
2. Please provide a practical explanation of File Content Manipulation
3. How to move and rename files?
4. Command to remove files and directories?

Solution by the Instructor:

1. Please provide a practical explanation of Basic File and Directory Operations.
    i. Open the terminal.
    ii. Use the **mkdir** command to create a directory named **lab_files**.
    iii. Navigate into the **lab_files** directory using the **cd** command.
    iv. Inside **lab_files**, create three files named **file1.txt**, **file2.txt**, and **file3.txt** using the **touch** command.
    v. List the contents of the current directory using the **ls** command.
    vi. Use the **cp** command to make a copy of **file1.txt** named **file1_copy.txt**.

2. Please provide a practical explanation of File Content Manipulation
    i. Open **file2.txt** using a text editor of your choice. Add some text to the file and save the changes.
    ii. Use the **cat** command to display the contents of **file2.txt**.
    iii. Append additional text to **file2.txt** using the **echo** command.
    iv. Create a new file named **merged_files.txt** that contains the concatenated contents of **file1.txt** and **file2.txt**.

3. How to move and rename files?
    i. Move **file3.txt** to the parent directory (**lab_files**) using the **mv** command.
    ii. List the contents of both the current directory and the parent directory to confirm the move.

iii. Rename **file3.txt** to **renamed_file.txt** using the **mv** command.
iv. List the contents again to verify the rename operation.

4. Command to remove files and directories?
    i. Navigate back to the home directory using the **cd** command.
    ii. Delete the entire **lab_files** directory and its contents using the **rm** command with appropriate options.
    iii. Confirm the deletion by listing the contents of the home directory.